Estimating Implicit Regularization in Deep Learning

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What regularization is implicit in your training procedure?

Motivation. Explicit regularization adds penalties to the training objective, steering learning toward solutions such as smaller weights to reduce overfitting and improve generalization. Early stopping, minibatching, and dropout also bias which solutions are learned, but their effects are harder to characterize [1, 2].

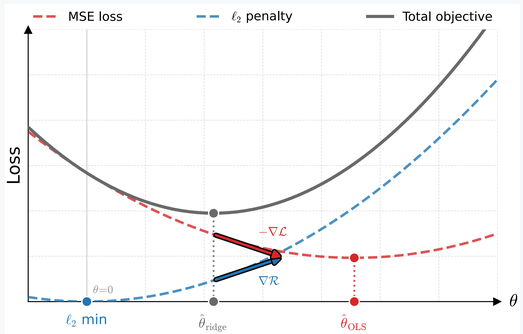
The gap. Theory explains some implicit effects, mostly in simplified settings [3, 4]. How can we test these predictions and study networks beyond analytical reach?

Work backwards. Rather than choosing a penalty and learning weights, observe the weights or training dynamics and infer a penalty that explains them. We choose a regularizer family and estimate its parameters by gradient matching—an empirical bridge between theory and practice.

Gradient matching

Training minimizes empirical risk (ERM). We model the learned weights as a regularized ERM (RERM) solution, choosing a family $\mathcal{R}(\theta, D; \Lambda)$ and estimating its parameters Λ:

$$\hat{\theta}_{\text{ERM}} = \arg\min_{\theta} \mathcal{L}(\theta, D) \qquad \longrightarrow \qquad \hat{\theta}_{\text{RERM}} = \arg\min_{\theta} \left\{ \mathcal{L}(\theta, D) + \mathcal{R}(\theta, D; \Lambda) \right\}$$



An ℓ_2 penalty shifts the solution, where the empirical loss gradient and regularizer gradient cancel.

Estimating from learned weights

At a hypothetical optimum $\hat{\theta}$ of $\mathcal{L} + \mathcal{R}$, the gradients sum to zero:

$$-\nabla_\theta \mathcal{L}(\hat{\theta}, D) = \nabla_\theta \mathcal{R}(\hat{\theta}, D; \Lambda)$$

These $p = \dim(\theta)$ equations let us solve for, or estimate, Λ:

$$\hat{\Lambda} = \arg\min_{\Lambda} \frac{1}{p} \|\nabla_\theta \mathcal{L} + \nabla_\theta \mathcal{R}_\Lambda\|_2^2$$

Example: For predictions $\hat{y} = f_\theta(X)$, MSE $\mathcal{L} = \|y - \hat{y}\|^2/n$ and $\mathcal{R} = \lambda\|\theta\|^2$ give

$$J^\top(y - \hat{y})/n = \lambda\hat{\theta}$$

where $J = \partial\hat{y}/\partial\theta$ is the prediction Jacobian, evaluated at $\hat{\theta}$.

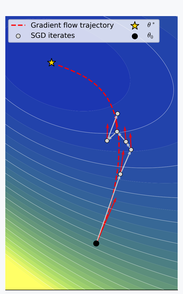
Estimating from weight updates

SGD iterates deviate from the full-batch gradient-flow path (right). For a gradient-step reference, match the update residual:

$$-\frac{\Delta\theta_t}{\eta} - \nabla_\theta \mathcal{L}(\theta_t) \;\approx\; \nabla_\theta \mathcal{R}(\theta_t; \Lambda)$$

where $\Delta\theta_t = \theta_{t+1} - \theta_t$.

Pool steps to fit a shared regularizer, or let its parameters vary over time. A zero update recovers the endpoint case.



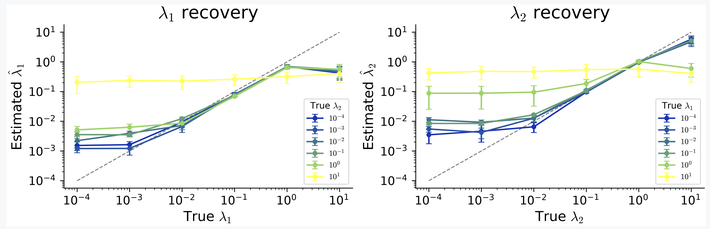
Recovering known regularization

Two ways to know that we have successfully estimated the implicit regularization:

Recover the penalty: match known regularization parameters.
Recover the weights: retrain with the estimated penalty to reproduce the learned weights.

Recovering elastic-net coefficients

Estimate both λ_1 and λ_2 in $\mathcal{R} = \lambda_1\|\theta\|_{1,\beta} + \lambda_2\|\theta\|_2^2$.



Deep ReLU networks; 6×6 penalty grid, 10 datasets/cell. Mean ± SE; dashed line: exact recovery.

Recovery is best at intermediate strengths. For $|\theta_j| < \beta = 10^{-3}$, smoothed ℓ_1 is quadratic, $\theta_j^2/(2\beta)$: its gradient is proportional to the ℓ_2 gradient. Strong penalties make the two effects hard to distinguish; weak penalties give little signal.

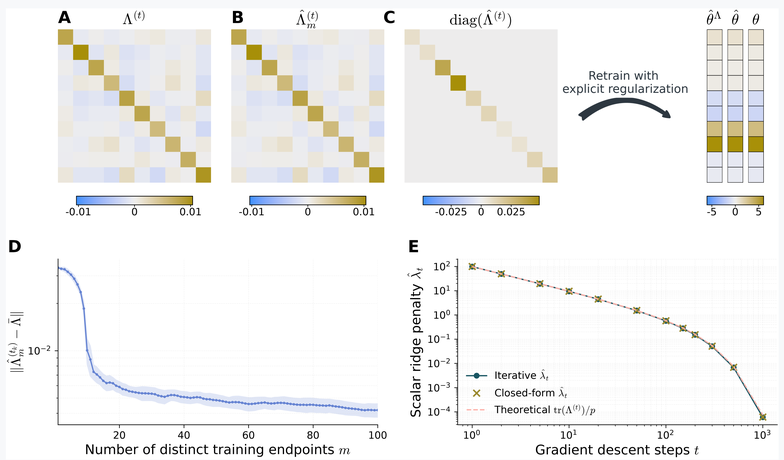
Early stopping as a quadratic penalty

The early-stopped OLS iterate minimizes $\|y - X\theta\|_2^2/n + \mathcal{R}_t(\theta)$ [5]:

$$\mathcal{R}_t(\theta) = \theta^\top \Lambda^{(t)} \theta, \qquad \Lambda^{(t)} = VS\left[(I - \eta S)^{-t} - I\right]^{-1} V^\top$$

Here $X^\top X/n = VSV^\top$, $\theta_0 = 0$, and $0 < \eta s_i < 1$.

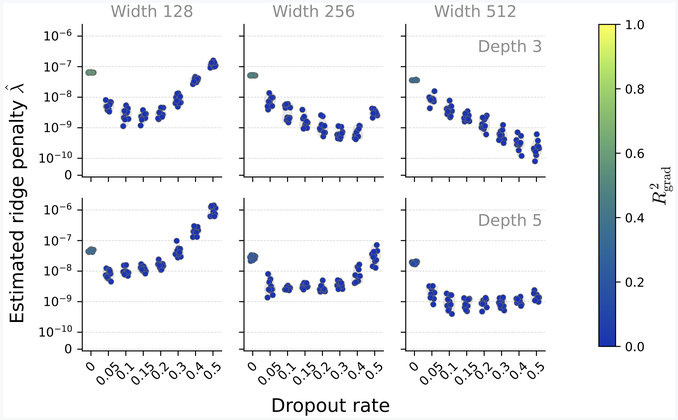
A symmetric Λ has $p(p+1)/2$ unknowns; each endpoint supplies p equations. For B, we pool diverse endpoints with shared X and t and varied responses. Diagonal fits use only p parameters.



A–B: Theoretical matrix (A) and estimate from 10 endpoints at a shared stopping time (B).
C: A diagonal fit from one endpoint differs from theory, yet retraining reproduces the early-stopped weights: an equivalent, underparameterized regularizer.
D: Matrix recovery improves with more endpoints, even when stopping times vary.
E: The estimated scalar penalty decays over training.

Implicit regularization in deep networks

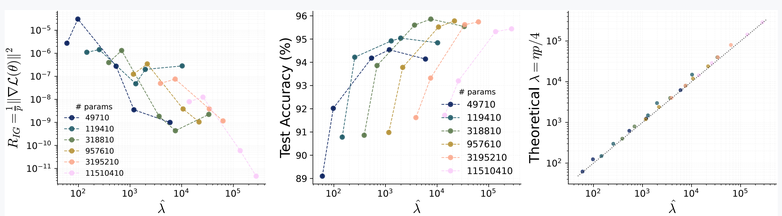
Dropout This training procedure has data-dependent regularization effects [6]. On MNIST, vary depth, width, and dropout rate; fit $\mathcal{R}(\theta, \lambda) = \lambda\|\theta\|_2^2$.



10 seeds per setting, including zero-dropout controls. Lighter color indicates better gradient fit.

In narrow networks, estimated ridge strength rises at high dropout; wider networks have smaller estimates. Ridge captures part of the effect.

Implicit gradient regularization Finite GD steps induce an approximate penalty $\mathcal{R} = \frac{\lambda}{p}\|\nabla\mathcal{L}\|^2$ with strength $\lambda = \eta p/4$ [7]. Penalizing step trajectories favors flatter minima associated with generalization. We recover λ (right); stronger regularization accompanies smaller gradients and higher test accuracy (left, middle).



Future work

- **Diagnose training choices early**, before expensive full runs.
- **Study complex architectures and optimizers** through their implicit penalties.
- **Improve identification** with diverse endpoints and richer regularizer families.

References

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